

CV1003-1– ENGINEERING MECHANICS
(For CV)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. Which of the following is not a basic idealization of Engineering Mechanics
A) Self-Weight B) Particle
C) Continuum D) Rigid Body
2. The S.I. unit of Force is
A) Kilo gram B) Newton
C) Kilowatt D) Calorie
3. Forces are said to be concurrent when they are acting
A) In same planes B) At a point
C) In different planes D) None of the above
4. A single force which is capable of producing the same effect as that of system of forces on a body is called as
A) Resultant B) Equilibrant
C) Couple D) Point force
5. The point through which the whole weight of the body acts is called
A) Central point B) Centroid
C) Inertial point D) Center of gravity
6. Where will be the center of gravity of a uniform rod lies?
A) At its center of its cross-sectional area B) At its end
C) At its middle point D) Depends upon its material
7. Which of the following laminas do not have centroid at its geometrical center
A) Right angled triangle B) Equilateral triangle
C) Circle D) Square
8. What is the moment of inertia of a semi-circular section of radius R about its centroidal axis parallel to diameter?
A) $\pi R^4/8$ B) $0.055 R^4$
C) $0.11 R^4$ D) $\pi R^4/16$
9. The friction experienced by a body when it is in equilibrium is known as
A) Sliding Friction B) Static Friction
C) Dynamic Friction D) Rolling Friction
10. The magnitude of limiting friction bears a constant ratio to is called coefficient of friction
A) Dry friction B) Frictional Resistance
C) Wet Friction D) Normal Reaction
11. The angle made by the Resultant reaction with Normal Reaction called
A) Angle of Friction B) Angle of Repose
C) Limiting Friction D) Cone of Friction
12. The maximum inclination of the plane with the horizontal on which a body free from external forces can rest without sliding is called
A) Angle of Friction B) Cone of Friction
C) Angle of Repose D) Limiting Friction

13. Example for cantilever beam is _____
 A) Railway sleepers B) Roof slab
 C) Bridges D) Portico slabs
14. Hinge support is called as
 A) Ball joint B) Swivel joint
 C) Pin joint D) Socket joint
15. Roller support is same as
 A) Simply support B) Hinged support
 C) Fixed support D) Roller support
16. What is the bending moment at end supports of a simply supported beam?
 A) Minimum B) Zero
 C) Maximum D) Uniform
17. A cantilever beam loaded with U.D.L throughout, the maximum shear force occurs at
 A) At point of contra-flexure B) Free end
 C) Centre D) Fixed end
18. The shear force in a beam subjected to pure positive bending is _____
 A) Negative B) Positive
 C) Zero D) Can't decide
19. The rate of change of shear force is equal to _____
 A) Intensity of loading B) Direction of load
 C) Change in BMD D) Maximum bending
20. The bending moment diagram for a cantilever beam with point load, at the free end will be
 A) A parabolic curve B) A triangle with max. height under fixed end
 C) A triangle with max. height under free end D) None of the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Define force and explain its four characteristics.
 b) Explain four basic idealization of engineering mechanics.
 c) Locate the centroid of the shaded area about the axes shown in Fig. Q1(c).

Marks	BT*	CO*	PO*
4	L2	1	1,2
4	L2	1	1,2

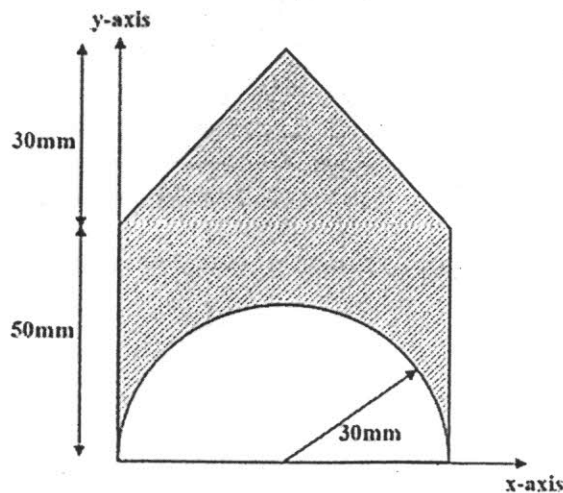


Fig. Q1(c)

2. a) Distinguish between centroid and center of gravity (any two point each).
 b) Derive the centroid of a rectangle of height 'h' and base 'b'.
 c) Determine the resultant of the coplanar non concurrent force system with respect to point 'C' as shown in Fig. Q2(c).

8	L3	2	1,2
4	L2	2	1,2
4	L2	2	1,2

